PHYS4055 Week 13 29/10/26

Q3: [sem. 2 2025 past exam]

a) Beam splitter, $c = (a+b)/\sqrt{2}$, $d = (a-b)/\sqrt{2}$

$$\frac{c+d}{\sqrt{2}} = a \quad , \quad \frac{c-d}{\sqrt{2}} = b$$

e.g. $|2,0\rangle = \frac{1}{\sqrt{2!}} a^{\dagger 2} |0,0\rangle$
$= \frac{1}{2\sqrt{2}} (c^\dagger + d^\dagger)^2 |0,0\rangle$

$$|\psi_{in}\rangle = \frac{1}{\sqrt{2}} (|2,0\rangle + |0,2\rangle)$$

$$|\psi_{out}\rangle = \frac{1}{\sqrt{2}} \left(\frac{1}{2\sqrt{2}} (c^{\dagger 2} + 2c^\dagger d^\dagger + d^{\dagger 2}) \frac{1}{2\sqrt{2}} (c^{\dagger 2} - 2c^\dagger d^\dagger + d^{\dagger 2}) \right) |0,0\rangle$$

$$= \tfrac{1}{4} \left(2(c^\dagger)^2 + 2(d^\dagger)^2 \right) |0,0\rangle$$

$$= \tfrac{1}{2} (c^{\dagger 2} + d^{\dagger 2}) |0,0\rangle$$

$$= \frac{\sqrt{2}}{2} \left(|2,0\rangle_{cd} + |0,2\rangle_{cd} \right)$$

, notation: $c^{\dagger 2} |0,0\rangle = \sqrt{2} |2,0\rangle_{cd}$

b) probabilities of output detections: P_{20}, P_{02}, P_{11}

$P_{20} = |\langle 2,0 | \psi_{out} \rangle|^2 = \left(\frac{\sqrt{2}}{2}\right)^2 = \tfrac{1}{2}$

$P_{02} = |\langle 0,2 | \psi_{out} \rangle|^2 = \left(\frac{\sqrt{2}}{2}\right)^2 = \tfrac{1}{2}$

$P_{11} = |\langle 1,1 | \psi_{out} \rangle|^2 = 0$

$\sum$ prob. $= \tfrac{1}{2} + \tfrac{1}{2} + 0 = 1$

c) 2 photons always exit together in the same port, never in separately both ports

↳ this is because interference (cancellation) of the $c^\dagger d^\dagger$ case (the cross term)

= related to/analogous to Hong-Ou-Mandel effect, where 2 indistinguishable photons give above result, 2 distinguishable photons do not

d) $|\Phi_{in}\rangle = \frac{1}{\sqrt{2}}(|1,1\rangle + |2,0\rangle)$, WTE P_{11}

$|1,1\rangle = a^\dagger b^\dagger |0,0\rangle = \frac{1}{2}(c^\dagger - d^\dagger)(c^\dagger + d^\dagger)|0,0\rangle$

$= \frac{1}{\sqrt{2}}(|2,0\rangle_{cd} - |0,2\rangle_{cd})$

$|2,0\rangle = \frac{1}{2}(|2,0\rangle_{cd} + \sqrt{2}|1,1\rangle_{cd} + |0,2\rangle_{cd})$

$|\Phi_{out}\rangle = \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}|2,0\rangle_{cd} - \frac{1}{\sqrt{2}}|0,2\rangle_{cd} + \frac{1}{2}|2,0\rangle_{cd} + \frac{\sqrt{2}}{2}|1,1\rangle_{cd} + \frac{1}{2}|0,2\rangle_{cd}\right)$

$= \frac{1}{\sqrt{2}}\left(\frac{\sqrt{2}+1}{2}|2,0\rangle_{cd} + \frac{1-\sqrt{2}}{2}|0,2\rangle_{cd} + \frac{\sqrt{2}}{2}|1,1\rangle_{cd}\right)$

$= \frac{\sqrt{2}+1}{2\sqrt{2}}|2,0\rangle_{cd} + \frac{1-\sqrt{2}}{2\sqrt{2}}|0,2\rangle_{cd} + \frac{1}{2}|1,1\rangle_{cd}$

$\therefore P_{11} = |\langle 1,1|\Phi_{out}\rangle|^2$, $P_{20} = |\langle 2,0|\Phi_{out}\rangle|^2$ 0.73

$= 1/4 = 0.25$ $= \frac{3+2\sqrt{2}}{8} \times$ [redacted].

$P_{02} = \frac{3-2\sqrt{2}}{8} \times 0.02$

$\sum \text{prob} = 1 = 1/4 + 6/8$

Q4.

a) WTS: $\dot{a} = -\chi a^\dagger - \gamma/2 \cdot a + \sqrt{\gamma} A_{in}$, relates to W5L2

Damped-cavity: $\dot{a} = \frac{-i}{\hbar}[a, H_{sys}] - \frac{\gamma}{2}a + \sqrt{\gamma}A_{in}$

$H_i = i\hbar\frac{\chi}{2}(a^2 - a^{\dagger 2})$, $\chi \in \mathbb{R}$

$[a, H_i] = i\hbar\frac{\chi}{2}[a, a^2 - a^{\dagger 2}]$

$= i\hbar\frac{\chi}{2}([a,a^2] - [a,a^{\dagger 2}])$

$[a,a] = 0$

$= -i\hbar\frac{\chi}{2}(a^\dagger[a,a^\dagger] + [a,a^\dagger]a^\dagger)$, Recall $[a,a^\dagger] = 1$

$= -i\hbar\frac{\chi}{2}(a^\dagger + a^\dagger)$

$= -i\hbar\chi a^\dagger$

$$\Rightarrow \dot{a} = \frac{-i}{\hbar}(-i\hbar\chi a^\dagger) - \frac{\gamma}{2} + \sqrt{2\gamma}\,A_{in}$$

$$= -\chi a^\dagger - \gamma/2\, a + \sqrt{\gamma}\,A_{in}$$

(skip : 4 b), c), d) : not so ~~relevant~~ & relevant)

e) maximally entangled qubits

Example of a Bell state : $|\Phi\rangle = 1/\sqrt{2}(|00\rangle + |11\rangle)$

by any measurement (in the computational basis) give 00, 11

$P_{00} = |\langle 00|\Phi\rangle|^2 = 1/2$ (where $P_{01} = P_{10} = 0$)

$P_{11} = |\langle 11|\Phi\rangle|^2 = 1/2$

∴ the states are max. entangled

(density matrix ρ being maximally mixed : $\rho = \mathbb{I}/2$)

$\rho = 1/2\,(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|)$

↳ therefore, no other state correlated

∴ no other state involved

∴ secret (bit string)